

NecroMerger Simulation Project

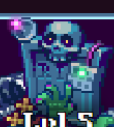
1 Description

This simulation is based off of the videogame NecroMerger, developed by Grumpy Rhino Games. The core gameplay loop involves summoning creatures and feeding them to level up the Devourer. Different stations summon different creatures, and creatures can be merged together to increase their level. Merging creatures contributes to spawning their corresponding champion; defeating champions give runes, a form of currency used throughout the entirety of the game.

2 Objective

2.1 Normal vs Hacked Stations

Stations have varying chances of producing different leveled creatures depending on the level of the station. As one progresses through the game, the ability to unlock hacked stations becomes available, increasing the cost of using the station in exchange for better odds. The figure below shows the information for a normal Grave on the left and a hacked Grave on the right.

 Lvl 1 Cost -250  100%	
 Lvl 2 Cost -300  60%  40%	 Lvl 2 Cost -400  100%
 Lvl 3 Cost -400  40%  30%  30%	
 Lvl 4 Cost -500  25%  25%  30%  20%	
 Lvl 5 Cost -700  10%  10%  40%  40%	 Lvl 5 Cost -1000  10%  90%

Is it more efficient to use hacked stations over normal ones?

2.2 Merging Methods

Consider the scenario shown below:



Figure 1: A level 4, 3, 2, 1, "0", and "-1" zombie. There is no more board space left to generate another creature.

Since merging higher level creatures contributes more progress towards spawning champions, merging high level creatures should be a priority. However, if there are no possible merges and no more space left to generate new creatures, what should one do? One option is to simply discard the highest level creature to make more space, so in this case the lv 4 zombie. Another option is to discard the flesh (lv "-1" zombie) in hopes that the station generates a hand (lv "0" zombie) so that a lv 5 zombie can be merged. However, this comes at the cost of potentially wasting many resources since a hand is not guaranteed to generate.

This leads to the following two merging methods if the board is full and cannot be merged. For both methods, any max leveled creature is discarded and any creatures that do not contribute to the specified champion are always discarded first.

1. High Discard: Discards the highest lv creature.
2. Low Discard: Discards the lowest lv creature if the station has a chance of generating a higher lv creature. Otherwise, discards the highest lv creature.

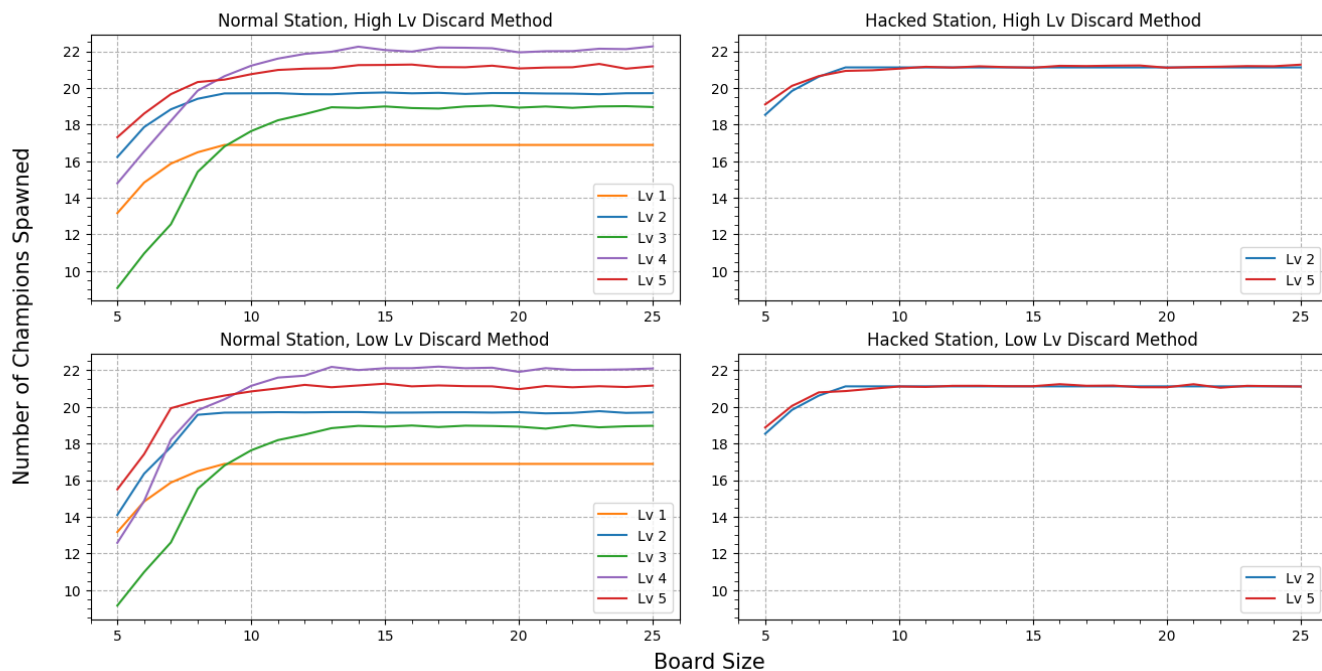
If there are multiple creature types on the board, the creature type that contributes fewer points towards spawning the champion is discarded. Note that discarding the "fewer points" creature is only relevant for the Grave since it is the only station that generates multiple creatures that contribute to spawning the same champion.

Does the high discard method spawn more champions than the low discard method? Is there even a significant difference? How does the amount of available board space affect the results?

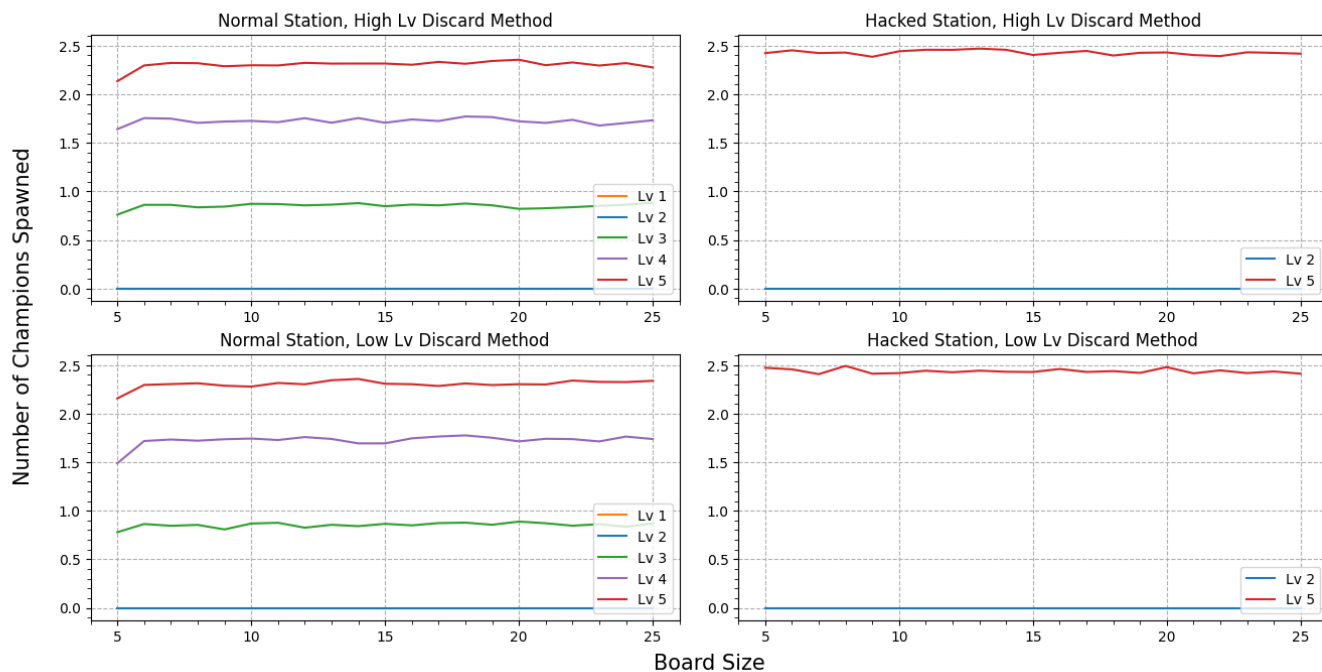
3 Data

3.1 Graphs

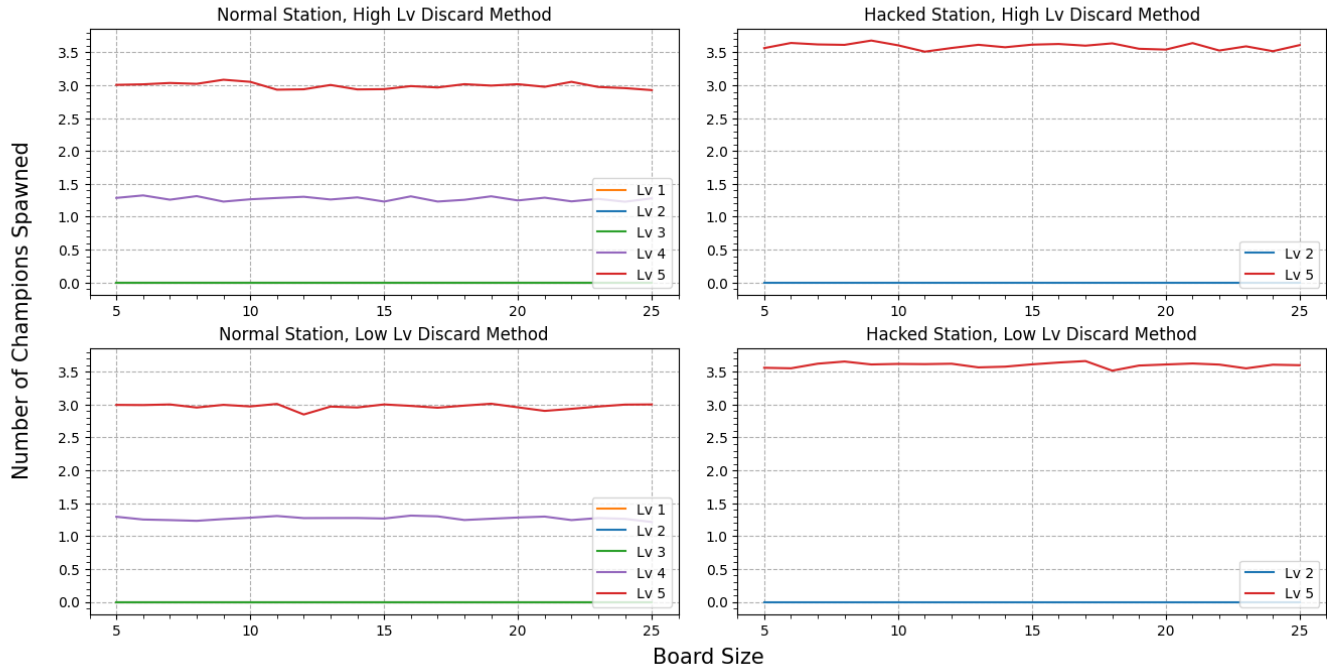
of Peasants spawned using Grave with $1.0e+06$ resources per trial



of Paladins spawned using Lectern with $1.0e+06$ resources per trial



of Rivals spawned using Portal with 1.0e+06 resources per trial



Each data point in the graphs shown was the average over 10 trials. Note that despite averaging over 10 trials, there is still some slight variance in the data. 100% champion spawning speed and 0% skip/duplicate chance were used. Champion spawning was simplified in the simulation to not take into account the tile used up when spawned (they effectively have their own tile and are immediately discarded).

3.2 Results

1. For spawning the Paladin and Rival, using the Lectern and Portal, respectively, the merge method made no difference regardless of board size. The average difference over all board sizes was less than .02 champions for all levels, normal and hacked. For spawning the Peasant using the Grave, in the small board size regime (5-10), the high discard method resulted in an average of .76, .70, .41 more Peasants at normal lv 2, 4, 5, respectively, corresponding to ~ 19 vs ~ 18 Peasants. Otherwise, there was similarly no difference. **There is no significant difference between the two merging methods.**
2. At a **board size of 10 and above**, a **lv 4 normal Grave** spawns more Peasants than a lv 5 Grave, normal and hacked.
3. In the high board size regime (11-25), there is **no difference between a lv 5 normal vs hacked Grave** (both average to ~ 21.1). A **lv 5 hacked Lectern** results in an average of **5.7% more** Paladins than a lv 5 normal Lectern (2.43 vs 2.30). A **lv 5 hacked Portal** results in an average of **20.4% more** Rivals than a lv 5 normal Portal (3.60 vs 2.99).